

SAHIL CHAKRABORTY

AI Engineer

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SUMMARY

AI engineer shipping production LLM systems — agentic RAG, hybrid retrieval, evaluation-in-CI and guardrails — on a data-engineering foundation. Two live, evaluated agents; comfortable owning a feature from retrieval to deploy.

EXPERIENCE

Tata Consultancy Services, *Platform Engineer cum Data Engineer* 07/2025 – Present

- Ship features inside an enterprise GenAI/LLM framework — prompt orchestration and LLM response handling — and version production prompt templates against held-out tests.
- Build Python + SQL ETL with automated integration and ingestion (incremental loads, query optimisation) across the billing domain, on AWS (S3, EC2, IAM).
- Profile source systems for lineage, quality and PII exposure — the ingestion and redaction layer a RAG corpus needs.

LTIMindtree, *AI/ML Intern*

12/2023 – 01/2024

Fine-tuned FLAN-T5 with PEFT for text-to-SQL over ITR-6 filings; built the training corpus end to end — extraction, cleaning, entity tagging.

TECHNICAL SKILLS

Generative AI (RAG, hybrid retrieval (dense + BM25, RRF), reranking, embeddings & pgvector, LangGraph, MCP (client + server), prompt engineering, structured output & tool calling) | **LLMOps & Eval** (golden-set & trajectory evaluation, eval-in-CI, execution-accuracy, NDCG@5, MLflow tracing, guardrails (AST, prompt-injection), OWASP LLM & Agentic Top 10) | **ML & DL** (PyTorch, Hugging Face Transformers, PEFT / LoRA / QLoRA, FLAN-T5 fine-tuning, scikit-learn) | **Backend & Data** (Python (async, typing), FastAPI, SQLAlchemy 2 (async), Docker, Postgres + pgvector, Redis, SQL, Python ETL, AWS (S3 / EC2 / IAM), GitHub Actions)

PROJECTS

DataChat, *LangGraph · FastAPI · Postgres/pgvector · MLflow*

- Built a plan → retrieve → guardrail → execute → verify → explain agent; RAG-to-schema grounding via pgvector stops column hallucination.
- Enforced safety by construction — AST SQL guardrail + read-only least-privilege role; provider circuit breaker (Groq → Gemini → self-hosted Ollama).
- Gated every merge on a 26-case golden set: execution 0.810, refusal 1.00, SQL-valid 0.952, faithfulness 0.905 (temp 0), traced in MLflow.

Quorum, *Python · FastAPI · LangGraph · MCP · pgvector*

- Supervisor agent reviews PRs via the GitHub MCP server; hybrid retrieval (dense + BM25, RRF) grounds every finding, cite-or-drop; also ships as an MCP server.
- Measured retrieval NDCG@5 0.526; cut reranking after it cost -0.079 NDCG@5 at 63-91× latency; AST context-scoping trimmed tokens 34.86%.
- Trajectory eval on a 10-PR set: routing 100%, finding recall 0% (reported honestly), citation rate 1.00 by construction; durable human approval before any write.

Yardstick, *in progress — Python · scikit-learn · MLflow · LangGraph · FastAPI · Docker*

AutoML agent that trains tabular models and reports them honestly — leakage detection, trivial-baseline comparison, justified metric selection, MLflow registry + drift monitoring.

EDUCATION

KIIT, Bhubaneswar, *B.Tech, Computer Science & Engineering*

2021 – 2025

CGPA 8.89 / 10

CERTIFICATIONS

Microsoft Certified: Fabric Data Engineer Associate (Microsoft (Feb 2026)) |

Microsoft Certified: Azure Data Fundamentals (Microsoft (Dec 2025)) |

Generative AI: Introduction and Applications (IBM (Aug 2024)) | **Prompt Engineering for ChatGPT** (Vanderbilt University (Aug 2024))